

Decent Coaching Classes

IX CBSE

Mind Your Math - II

Answer Key: Test - Chapters 2, 3, 4 and 5 (up to 5.3)

SECTION A — MCQs

1. (C) $\sqrt{5}$
2. (B) 1
3. (A) $(a + b)(a - b)$
4. (B) circumcentre
5. (A) terminating [8 = 2^3 , prime factors of the denominator are only 2, so the decimal terminates: $7/8 = 0.875$]

SECTION B — Very Short Answer

6. $3(-2)^2 - 2(-2) + 5 = 3(4) + 4 + 5 = 12 + 4 + 5 = 21$
7. $40 = 2^3 \times 5$. Since the prime factors of the denominator are only 2 and 5, the decimal expansion of $17/40$ is terminating.
8. $998^2 = (1000 - 2)^2 = 1000^2 - 2(1000)(2) + 2^2 = 1000000 - 4000 + 4 = 996004$
9. (i) Infinitely many circles can pass through one given point. (ii) Infinitely many circles can pass through two given points — their centres all lie on the perpendicular bisector of the segment joining the two points.

SECTION C — Short Answer

10. Let $x = 1.2454545\ldots$ (one non-repeating digit '2', repeating block '45').
Multiply by 10 (to move the non-repeating digit before the decimal): $10x = 12.454545\ldots$...(i)
Multiply (i) by 100 (repeating block has 2 digits): $1000x = 1245.454545\ldots$...(ii)
Subtracting (i) from (ii): $990x = 1233 \Rightarrow x = 1233/990 = 137/110$.
11. Let the smaller number = x , larger = $76 - x$. Given: $2x = (76 - x) + 8 \Rightarrow 2x = 84 - x \Rightarrow 3x = 84 \Rightarrow x = 28$. The numbers are 28 and 48.
12. $4x^2 - 12xy + 9y^2 - 25z^2 = (2x - 3y)^2 - (5z)^2 = (2x - 3y - 5z)(2x - 3y + 5z)$
13. $PQ^2 + QR^2 = 5^2 + 12^2 = 25 + 144 = 169 = 13^2 = PR^2$, so $\triangle PQR$ is right-angled at Q. By the property of the circumcentre of a right-angled triangle, the circumcentre lies at the midpoint of the hypotenuse PR — that is, it lies ON the triangle.

SECTION D — Long Answer

14. Proof (by contradiction):
Assume $\sqrt{3}$ is rational. Then $\sqrt{3} = p/q$, where p and q are co-prime integers, $q \neq 0$.
Squaring: $3 = p^2/q^2 \Rightarrow p^2 = 3q^2$...(i)
So p^2 is divisible by 3, which means p is divisible by 3 (since 3 is prime). Let $p = 3k$ for some integer k .
Substituting in (i): $9k^2 = 3q^2 \Rightarrow q^2 = 3k^2$, so q^2 is divisible by 3, which means q is also divisible by 3.
Thus p and q have a common factor of 3, contradicting the assumption that p and q are co-prime.
Hence, our assumption is wrong, and $\sqrt{3}$ is an irrational number.